



Batterjee Medical College
For Science and Technology

Pulmonary Vascular Disease

An approach to a Case of Dyspnea

Dr. Essam Kotb
Associate professor,
Internal Medicine Department

Main Subtopics



- Pulmonary Embolism (PE)
- Pulmonary Hypertension

Pulmonary Embolism

Knowledge:

- Define and explain the underlying mechanisms of Pulmonary Embolism (PE), Deep Vein Thrombosis (DVT), and Pulmonary Hypertension (PH). Outline Virchow's Triad and identify risk factors for venous thromboembolism.
- Categorize Pulmonary Hypertension into its clinical groups, describe its genetic and environmental risk factors, and explain the development of secondary complications.
- Describe the typical signs, symptoms, and objective findings of PE, DVT, and PH).

Skills:

- Interpret diagnostic modalities for pulmonary vascular diseases, including chest X-rays, ECGs, CT Pulmonary Angiography, V/Q scans, echocardiograms, and arterial blood gas results.
- Apply clinical decision tools (such as Wells Criteria for PE and DVT) to determine pretest probability and choose appropriate diagnostic algorithms.
- Select and apply appropriate management plans based on patient stability, including the initiation of anticoagulation (DOACs, Heparin, Warfarin), thrombolytics, or targeted vasoactive therapies, while evaluating absolute and relative contraindications.

Values:

- Demonstrate a professional attitude by maintaining strict adherence to clinical training workplace rules, safety guidelines, and regulations.
- Recognize contraindications for anticoagulation and invasive procedures to minimize patient harm and ensure evidence-based ethical care.

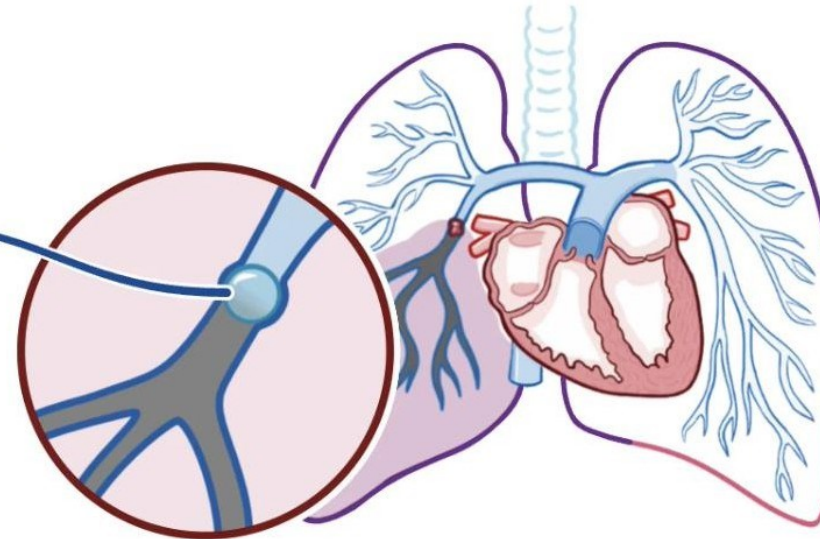
Pulmonary Embolism

Definition:

PULMONARY EMBOLISM

**BLOCKAGE of PULMONARY ARTERY
or BRANCHES by an EMBOLUS**

- BLOOD CLOT, TUMOR,
FRAGMENT of FAT, or AIR
that ORIGINATES from
SOMEWHERE ELSE in the BODY



Pulmonary Embolism

Etiology:

PE is **FATAL**

Fat

Air

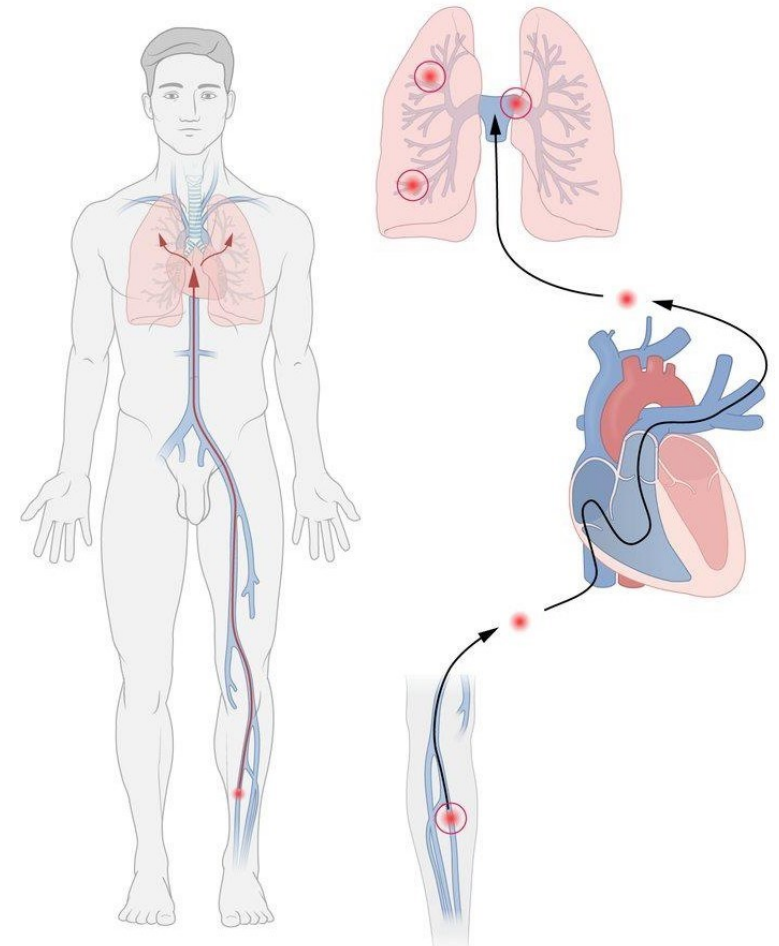
Thrombus

Amniotic fluid

Less common

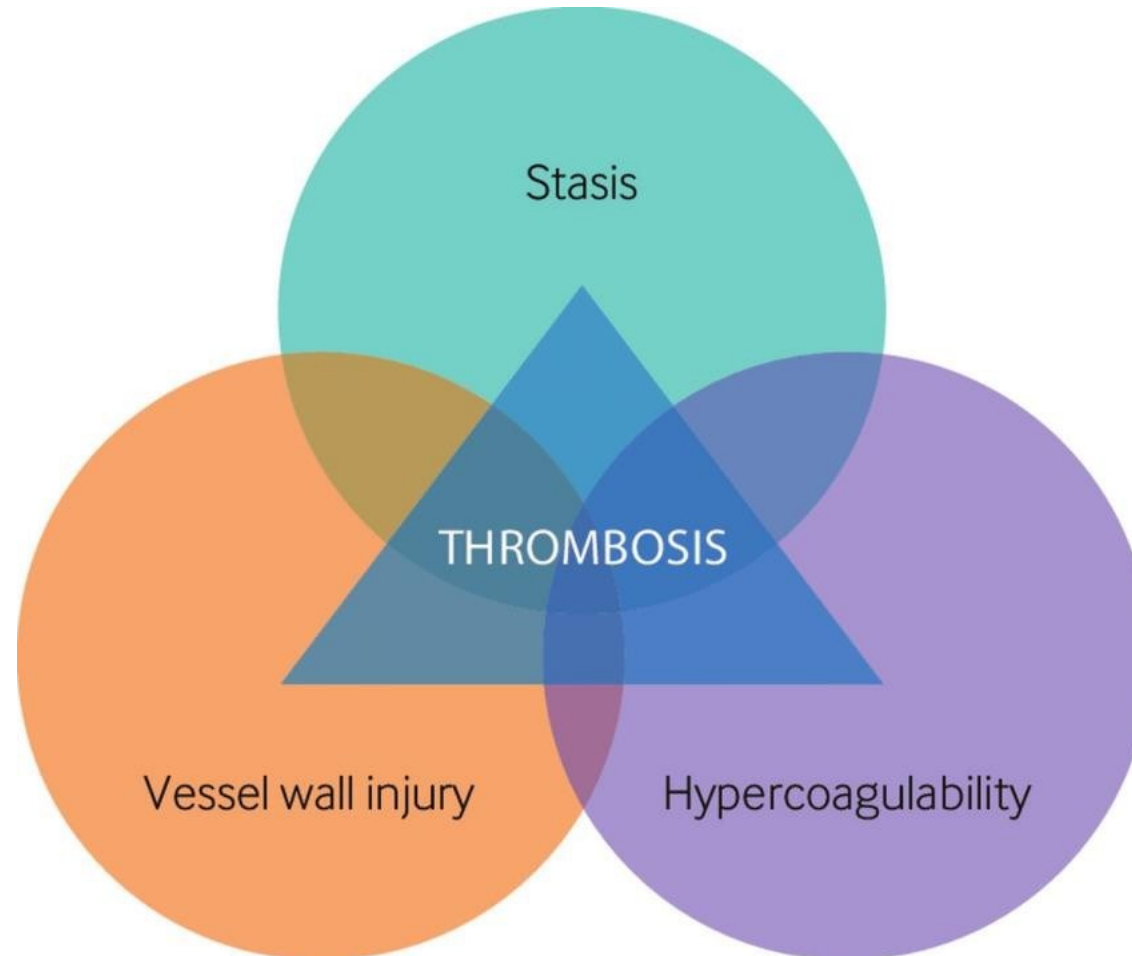
Most PEs arise from thromboses in the deep veins of lower extremities above the knee:

- Iliac
- Femoral
- Popliteal



Pulmonary Embolism

Predisposing Factors - Virchow's Triade: → DVT



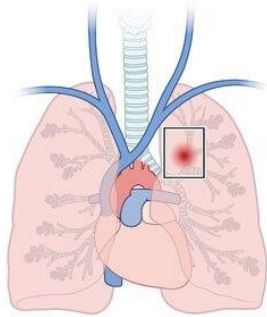
Pulmonary Embolism

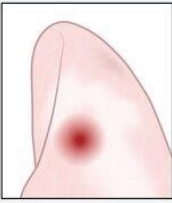
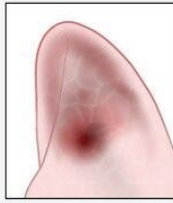
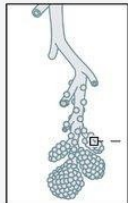
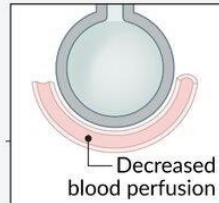
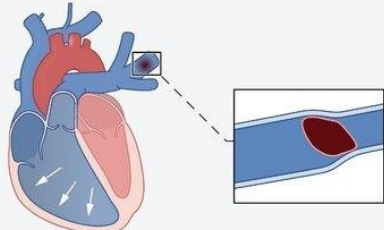
Risk Factors:

- Age >60 years
- Malignancy
- Prior history of DVT, PE
- Hereditary hypercoagulable states (factor V Leiden, protein C and S deficiency, antithrombin III deficiency)
- Prolonged immobilization or bed rest, long-distance travel
- Cardiac disease
- Obesity
- Nephrotic syndrome
- Major surgery, especially pelvic surgery (orthopedic procedures)
- Major trauma
- Pregnancy, estrogen use (oral contraceptives)

Pulmonary Embolism

Pathogenesis:



	Pathophysiology	Presentation
Local infarction	  Ischemia → Pleura and lung inflammation	 Fever  Hemoptysis  Pleuritic chest pain
Impaired gas exchange	  ↓ V:Q ↓ PaO ₂ ↑ A-a gradient ↓ PaCO ₂ Respiratory alkalosis	 Dyspnea  Cough  Wheezing
Cardiac compromise	 Right ventricular overload ↑ Pulmonary artery pressure ↓ Cardiac output	 Tachycardia  Shock/syncope

Pulmonary Embolism

Symptoms

- Common features of PE
- Acute onset of symptoms
- Dyspnea
- Tachycardia and tachypnea
- Sudden pleuritic chest pain
- Cough and hemoptysis
- Associated features of DVT

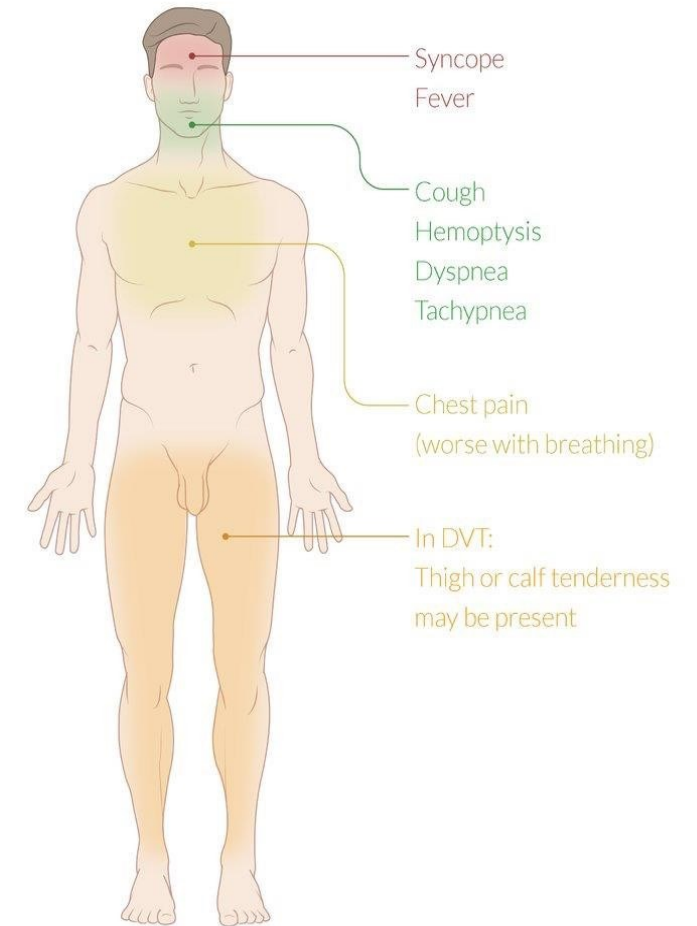
Less common features of PE

- Decreased breath sounds
- Dullness to percussion
- Split S2
- Low-grade fever

Features of massive PE

Presyncope or syncope

- Jugular venous distension; and Kussmaul sign
- Hypotension and obstructive shock
- Circulatory collapse

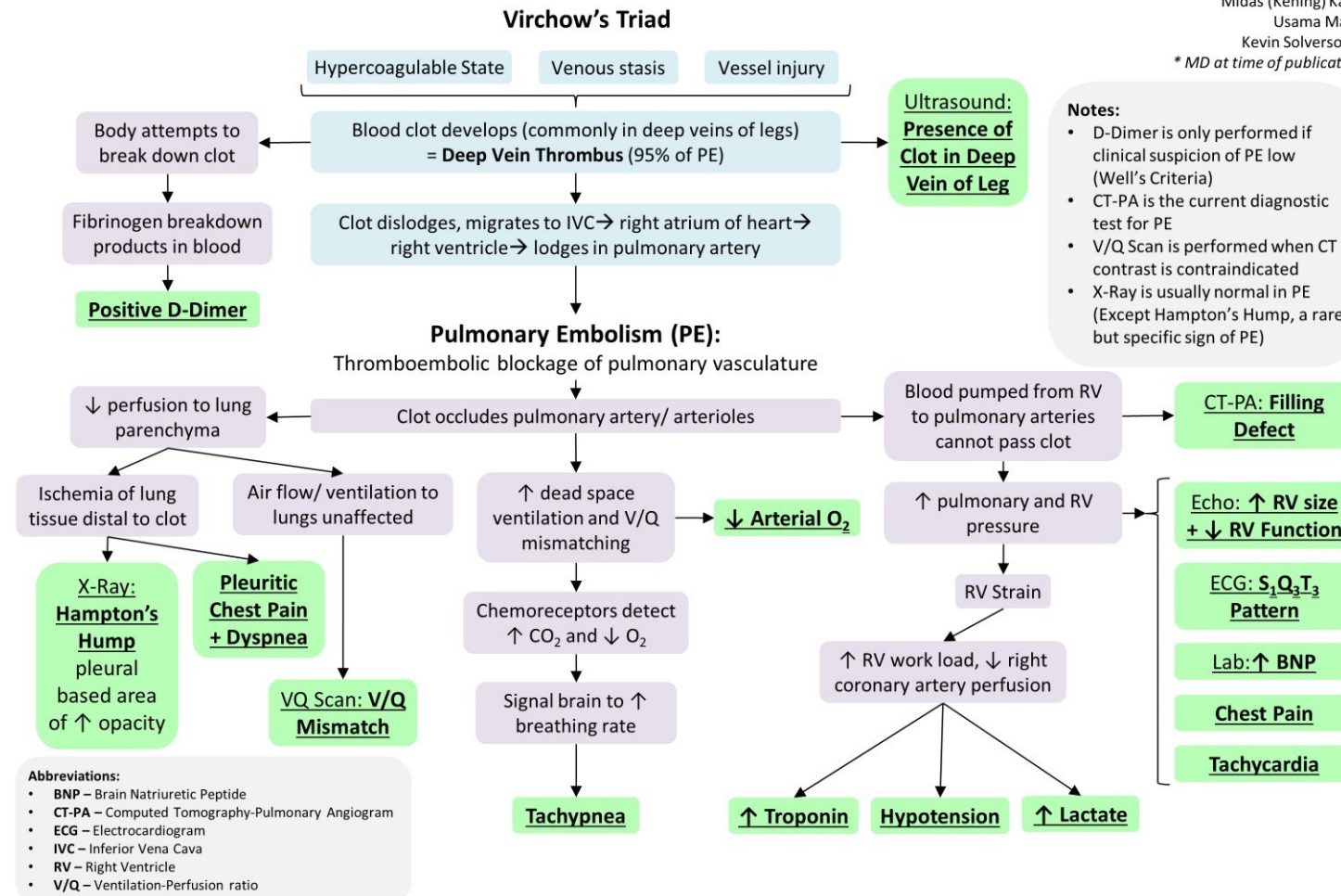


Pulmonary Embolism

Diagnostics:

Pulmonary Embolism: Pathogenesis and Laboratory Findings

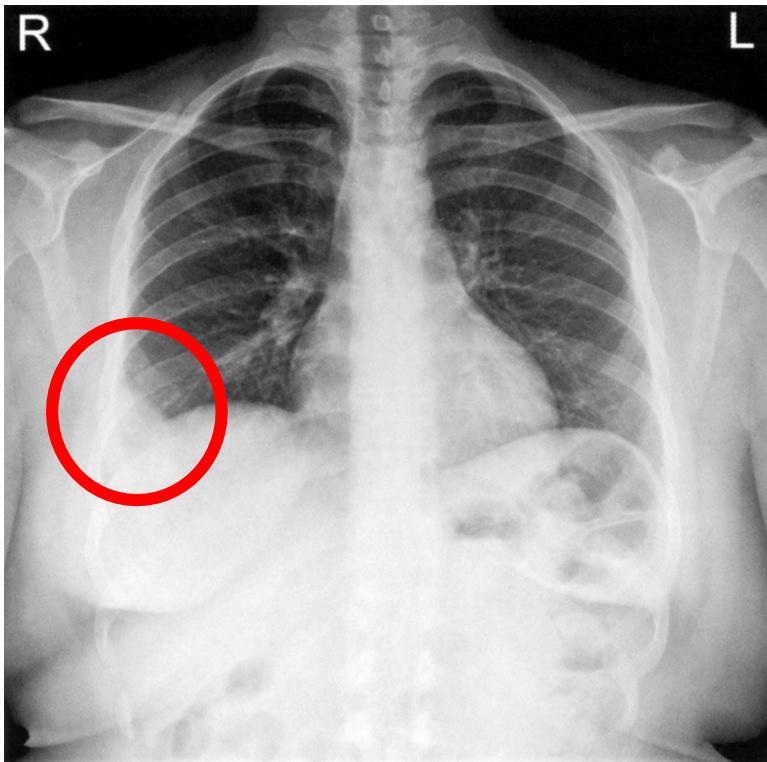
Authors:
Mackenzie Gault
Reviewers:
Midas (Kening) Kang
Usama Malik
Kevin Solverson *
* MD at time of publication



Pulmonary Embolism

Diagnostics:

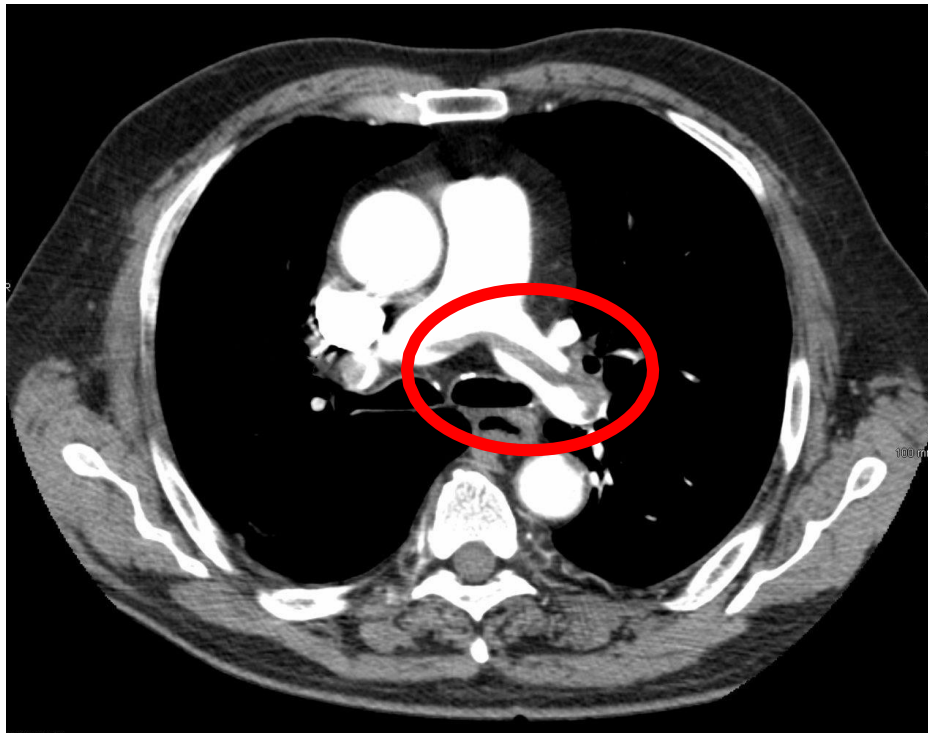
- CXR is usually normal.
- Classic radiographic signs, such as *Hampton hump* or *Westermark sign*, are rarely present.



Pulmonary Embolism

Diagnostics:

- CTA Has been found to have good sensitivity (>90%) and specificity.
- Intraluminal filling defects in central, segmental, or lobular pulmonary arteries on CTA
- CTA cannot be performed in patients with significant renal insufficiency

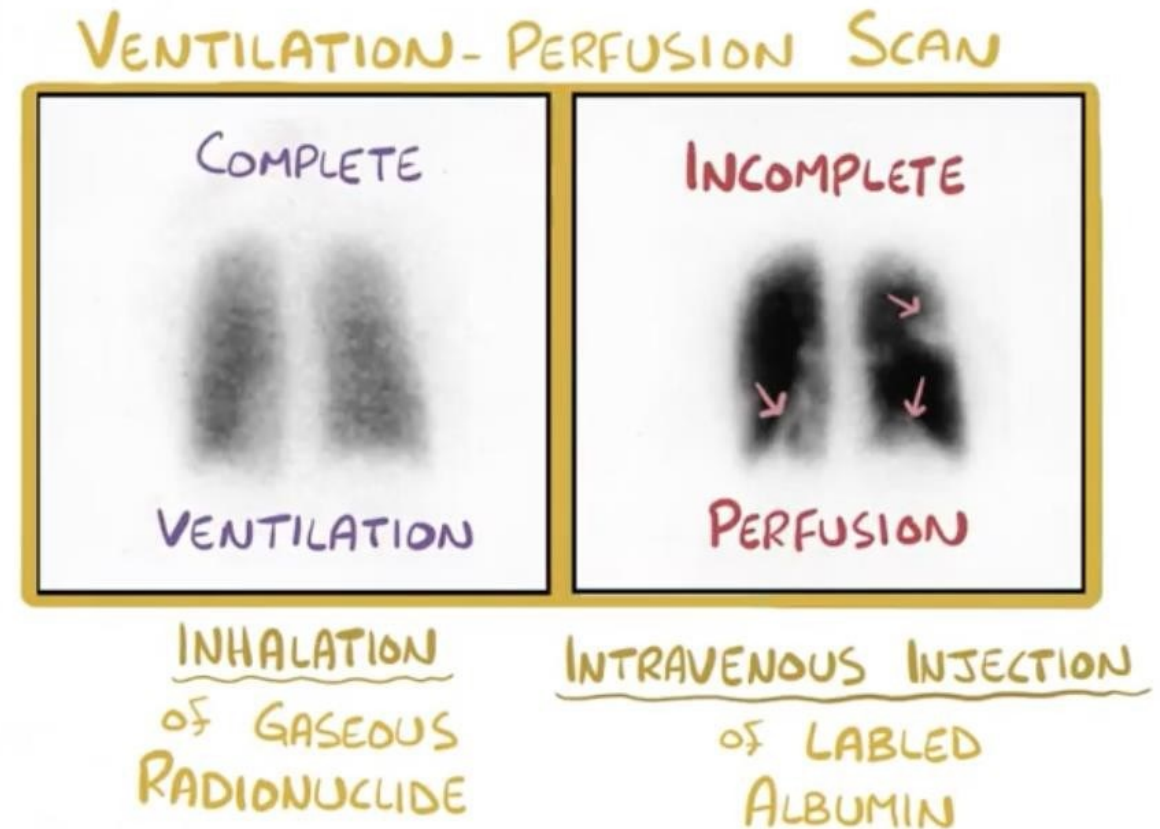


Pulmonary Embolism

Diagnostics:

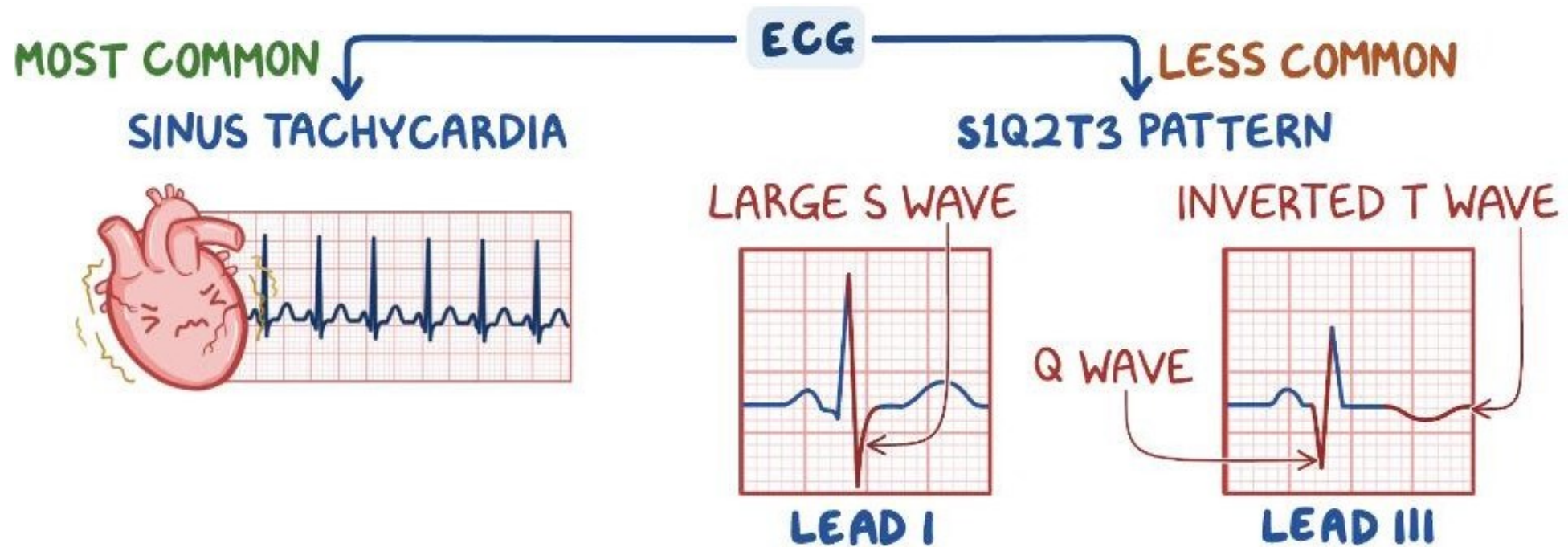
- V/Q scan Plays an important role in diagnosis when there is a contraindication to CTA.

History of previous anaphylactic reactions,
Acute kidney injury
Severe chronic kidney disease,
Hyperthyroidism.



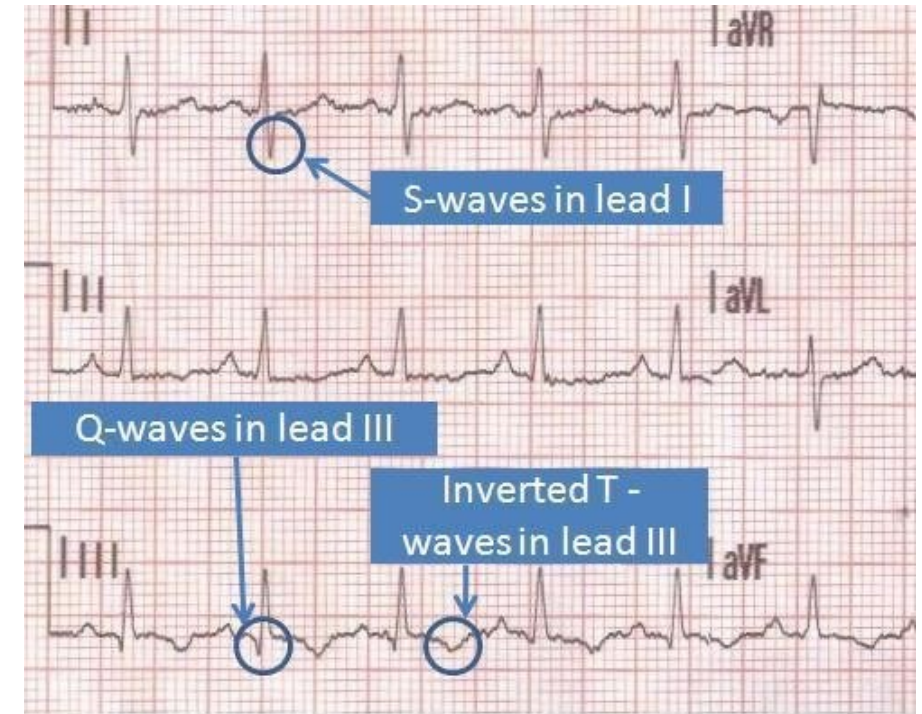
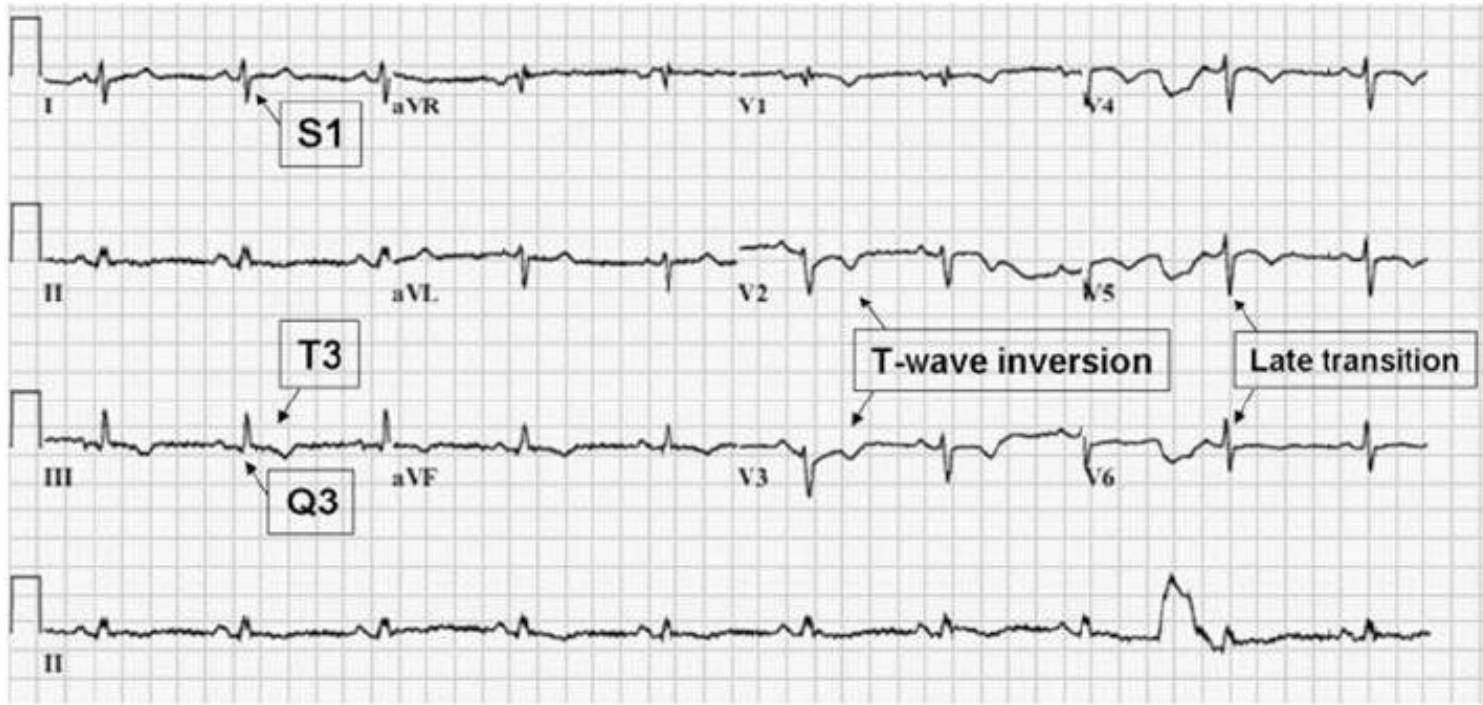
Pulmonary Embolism

Diagnostics:



Pulmonary Embolism

Diagnostics:



Pulmonary Embolism

Clinical Approach:

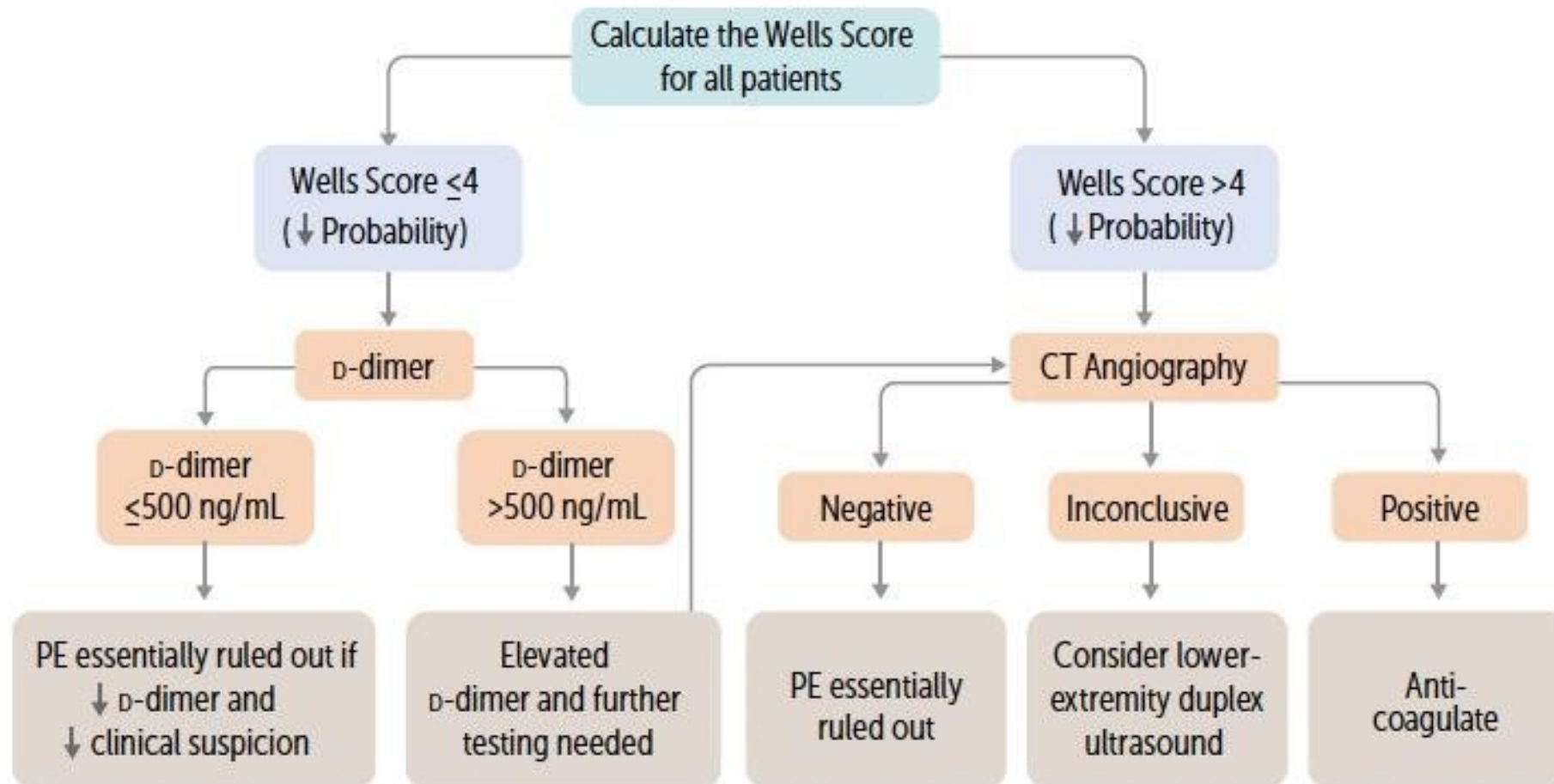
TABLE 10.1. Modified Wells Criteria for Pulmonary Embolism^a

CLINICAL FINDINGS THAT INCREASE RISK	POINTS
Clinical symptoms of DVT (leg swelling, pain with palpation)	3.0
Other diagnosis less likely than PE	3.0
Heart rate >100 bpm	1.5
Immobilization (≥ 3 days) or surgery in the previous 4 weeks	1.5
Previous DVT/PE	1.5
Hemoptysis	1.0
Malignancy	1.0

^aScores >4: PE is likely; scores ≤ 4 : PE is unlikely.

Pulmonary Embolism

Clinical Approach:



Pulmonary Embolism

Diagnostics in Pregnancy:

• Initial evaluation

- Use clinical suspicion to determine whether imaging is required.
- Avoid using the Wells score, as it is not validated for use in pregnancy or the postpartum period.

• Imaging

- Initial imaging: CXR
- Further imaging
 - If CXR is normal: ventilation/perfusion scintigraphy
 - If CXR is abnormal: CTPA

PE may be diagnosed based on clinical suspicion for PE and Doppler ultrasound findings of lower-extremity DVT.

Management of Pulmonary Embolism

- **Hemodynamically unstable patients:**

Begin management of **massive PE prior to diagnostic confirmation.**

- Assess for signs of RV dysfunction.
 - Obtain an ECG.
 - Perform a bedside echocardiogram or focused cardiac ultrasound
- Consider adding DVT point-of-care ultrasound
- Obtain confirmatory imaging (CTPA or V/Q scan) when stable enough for transport.

Pulmonary Embolism

Management of Unstable Patients:

- Patients with massive PE who are hemodynamically unstable (persistent hypotension).
- Patients with evidence of right heart failure (thrombolysis can reverse this).

! UNSTABLE

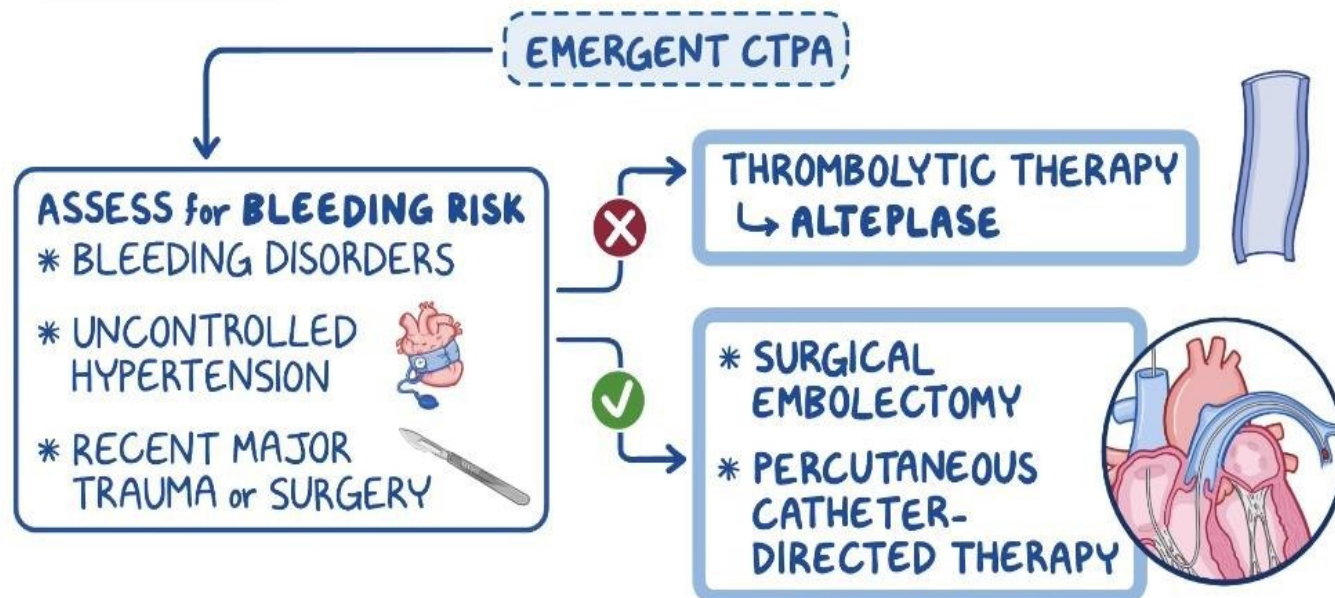
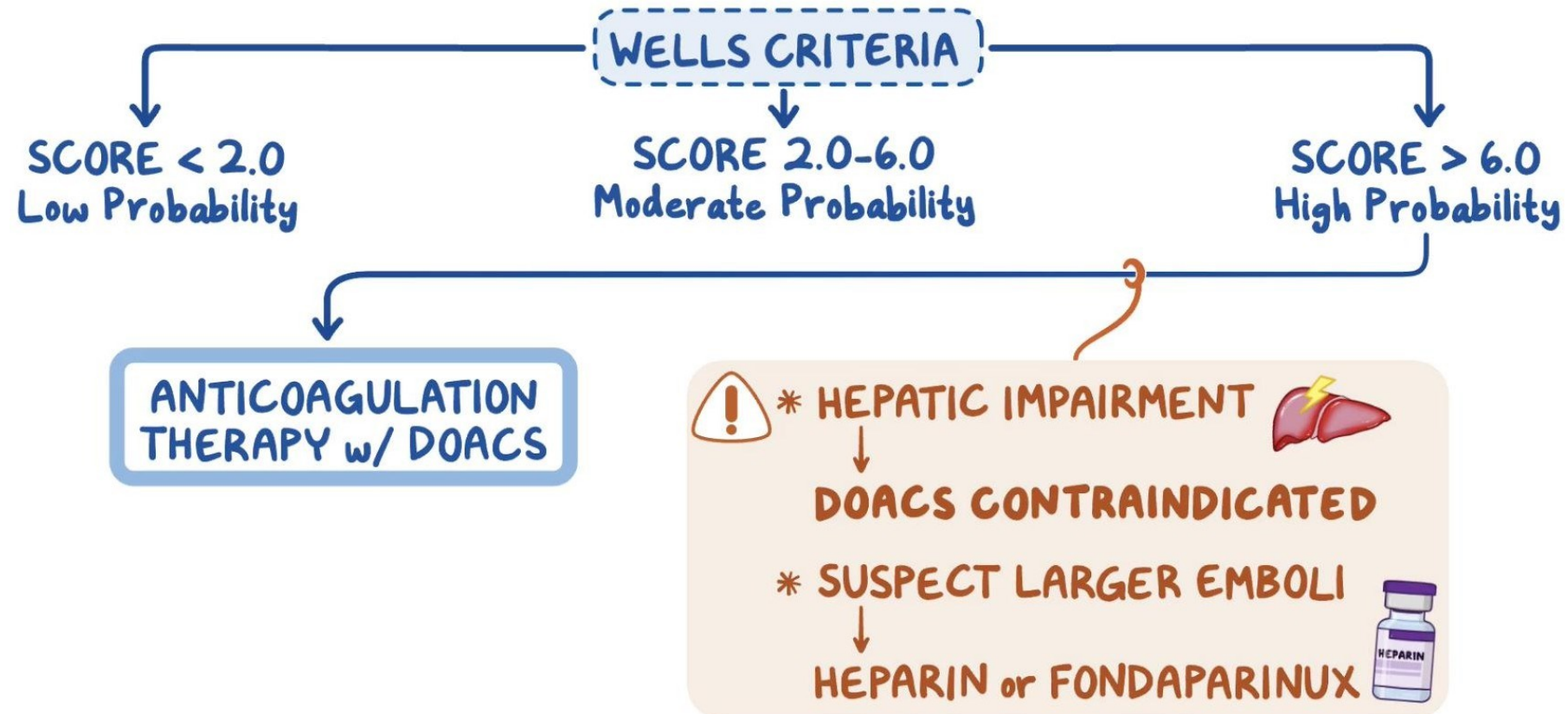


TABLE 10.4. Contraindications to All Anticoagulation

ABSOLUTE	RELATIVE
Hemorrhagic stroke	Recent internal bleeding (within 6 months)
Active internal bleeding	Prior hemorrhagic stroke
Suspected aortic dissection	Thrombocytopenia
	CNS mass lesion (especially RCC and melanoma)

Pulmonary Embolism

Management of Stable Patients:

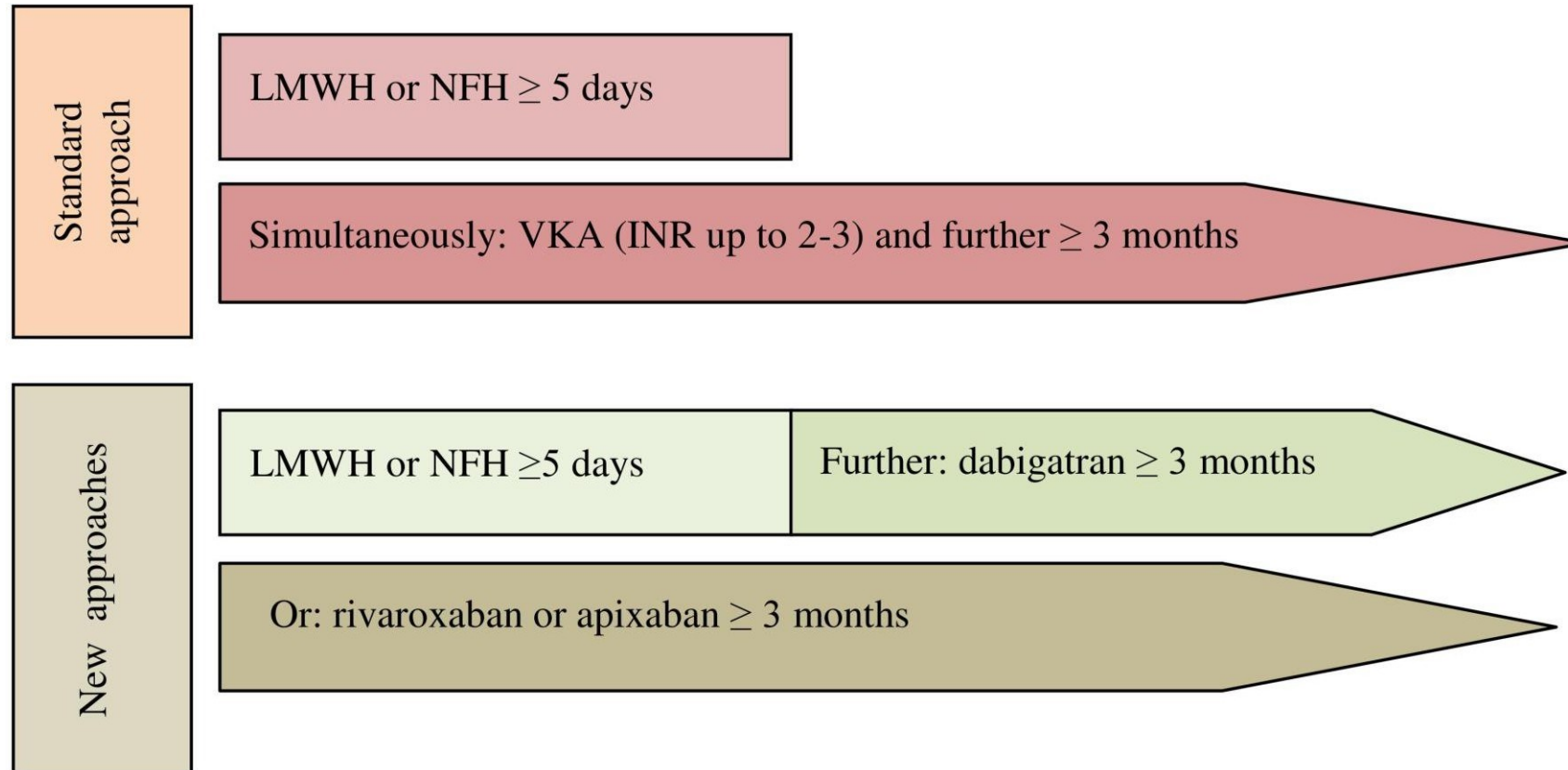


Pulmonary Embolism

- **Hemodynamically stable patients** (e.g., systolic BP > 90 mm Hg): Assess PTP of PE.
 - Low PTP of PE
 - **Pulmonary embolism rule-out criteria based on wells score = 0**: No further testing
 - **Pulmonary embolism rule-out criteria > 0**: Obtain D-dimer.
 - Elevated D-dimer: Obtain confirmatory imaging.
 - Normal D-dimer: No further testing, consider alternative diagnoses.
 - Intermediate PTP of PE: Obtain D-dimer.
 - Elevated D-dimer: Obtain confirmatory imaging.
 - Normal D-dimer: No further testing; consider alternative diagnoses.
 - High PTP of PE: Obtain confirmatory imaging.

Pulmonary Embolism

Management of Stable Patients:



Deep Vein Thrombosis

Clinical Features:

- Pain, swelling, or erythema of the affected extremity
- A palpable cord
- Low-grade fever
- Tender calf muscle



Deep Vein Thrombosis

Diagnostics:

- Duplex ultrasonography: DVT Wells Score >1
- d-dimer: DVT Wells score <1

Management:

- Similar to PE
- If contraindications for anticoagulation: IVC filter

TABLE 10.6. Wells Criteria for DVT

CLINICAL FINDINGS THAT INCREASE RISK	POINTS ^a
Active cancer	1
Paralysis, paresis, or recent plaster immobilization of the lower extremities	1
Recently bedridden for 3 days, major surgery in last 12 weeks requiring general or regional anesthesia	1
Localized tenderness along the deep venous system	1
Entire leg swollen	1
Calf swelling >3 cm compared to asymptomatic leg	1
Pitting edema confined to symptomatic leg	1
Non-varicose collateral superficial veins	1
Previously documented DVT	1
Alternative diagnosis at least as likely as DVT	-2

^aScores: <0 = low pretest probability; $1-2$ = moderate pretest probability; ≥ 3 = high pretest probability.

Pulmonary Hypertension

Definition:

Pulmonary hypertension is defined by an elevated mean pulmonary arterial pressure (mPAP) > 20 mm Hg at rest

Etiology:

Pulmonary arterial hypertension

Idiopathic, Hereditary

Drug-induced, Methamphetamine, Sympathomimetic appetite suppressants

Associated conditions

Connective tissue diseases

Congenital heart disease (Eisenmenger syndrome)

Left heart disease

Congestive heart failure

Valvular heart diseases, e.g. aortic valve stenosis, mitral valve stenosis

Chronic lung diseases

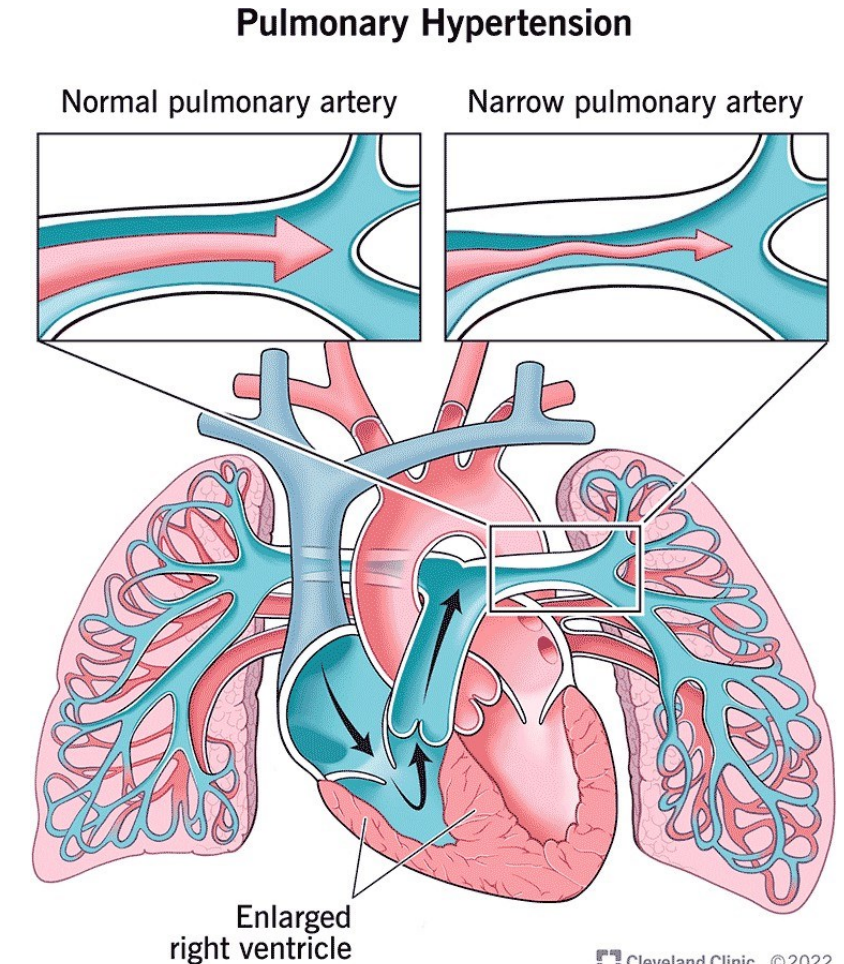
Obstructive lung diseases

Obstructive sleep apnea

Restrictive lung diseases

Pulmonary artery obstruction

Chronic thromboembolic



Pulmonary Hypertension

Pathophysiology:

Increased pulmonary vascular resistance

- Occlusive vasculopathy
- Hypoxic pulmonary vasoconstriction
- Increased pulmonary vessel pressure → right heart hypertrophy & failure
- Inflammation (e.g., COPD) → ↑ inflammatory cell infiltration of intima → intimal fibrosis
- Increased pulmonary venous pressure: volume or pressure overload from left-sided heart disease

Increased pulmonary blood flow

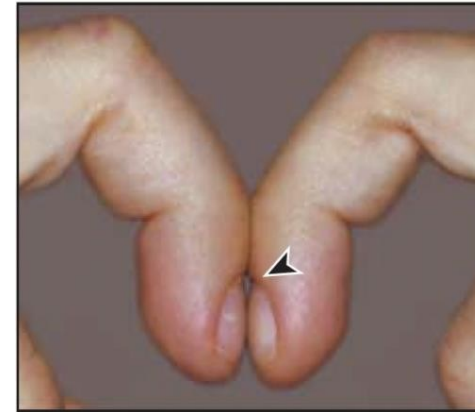
- Left-to-right shunt (e.g., ASD, VSD, PDA)
- Portopulmonary hypertension

Pulmonary Hypertension

Clinical Features:

- Dyspnea on exertion
- Fatigue
- Chest pain (exertional)
- Syncope (exertional)
- Clubbing
- Cyanosis
- Loud pulmonic component of the second heart sound (P2)
- Heave (sign of RV dilatation)
- Rt sided HF (JVD, hepatomegaly, ascites, peripheral edema)

Normal



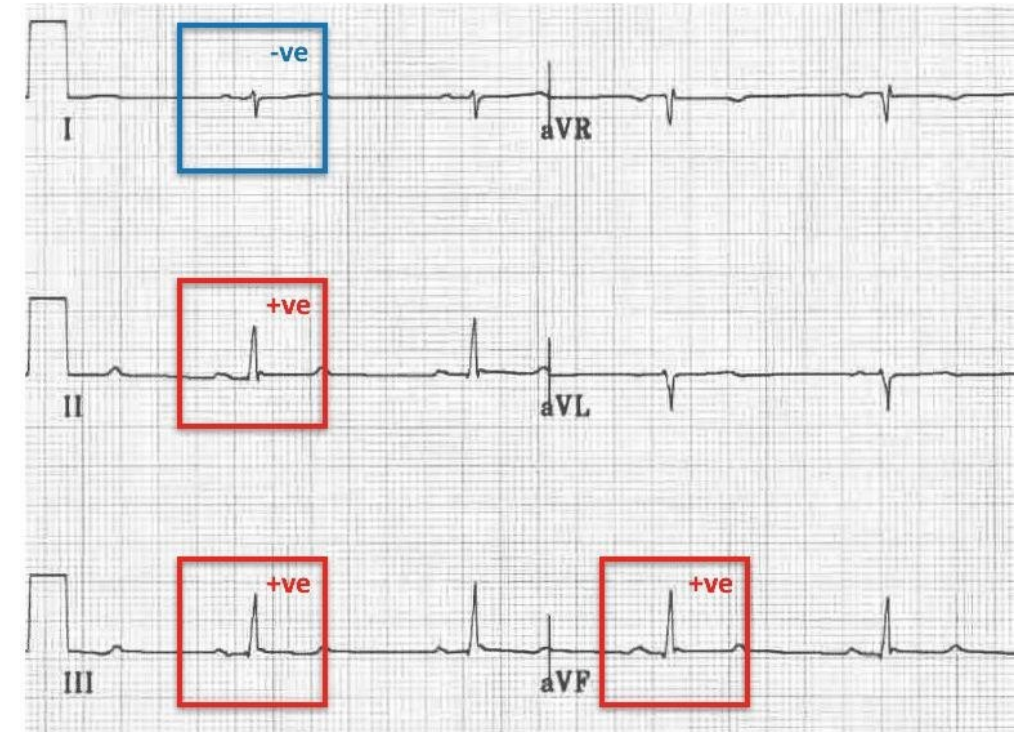
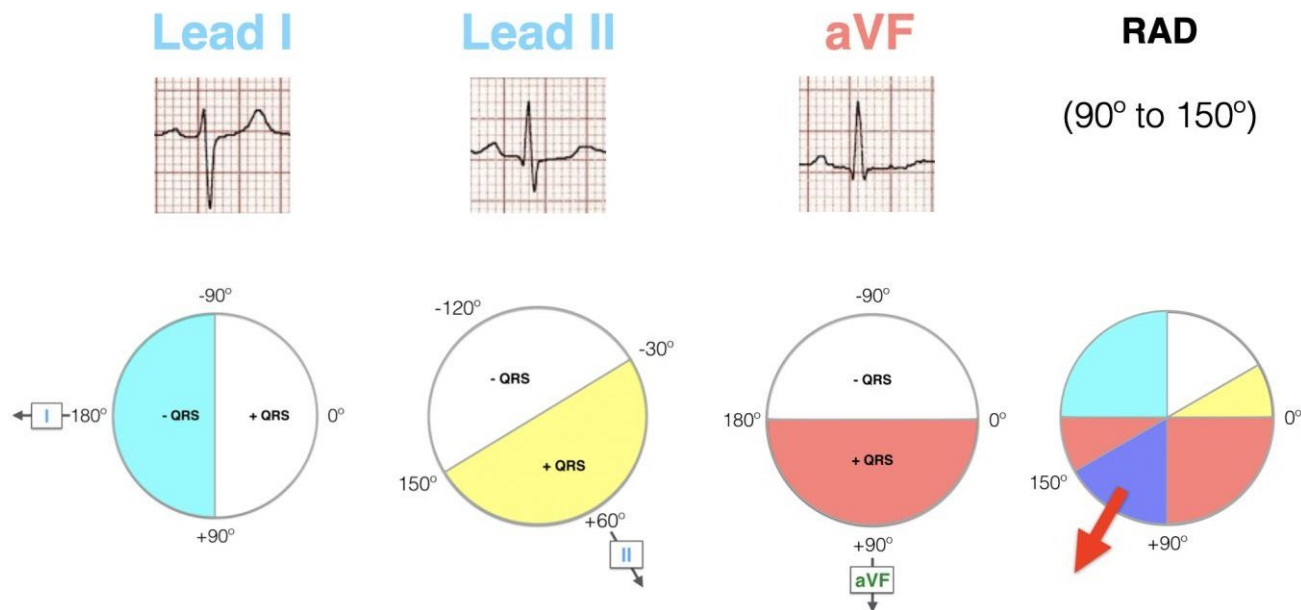
Clubbed



Pulmonary Hypertension

Diagnostics:

- ECG Often suggests right ventricular hypertrophy

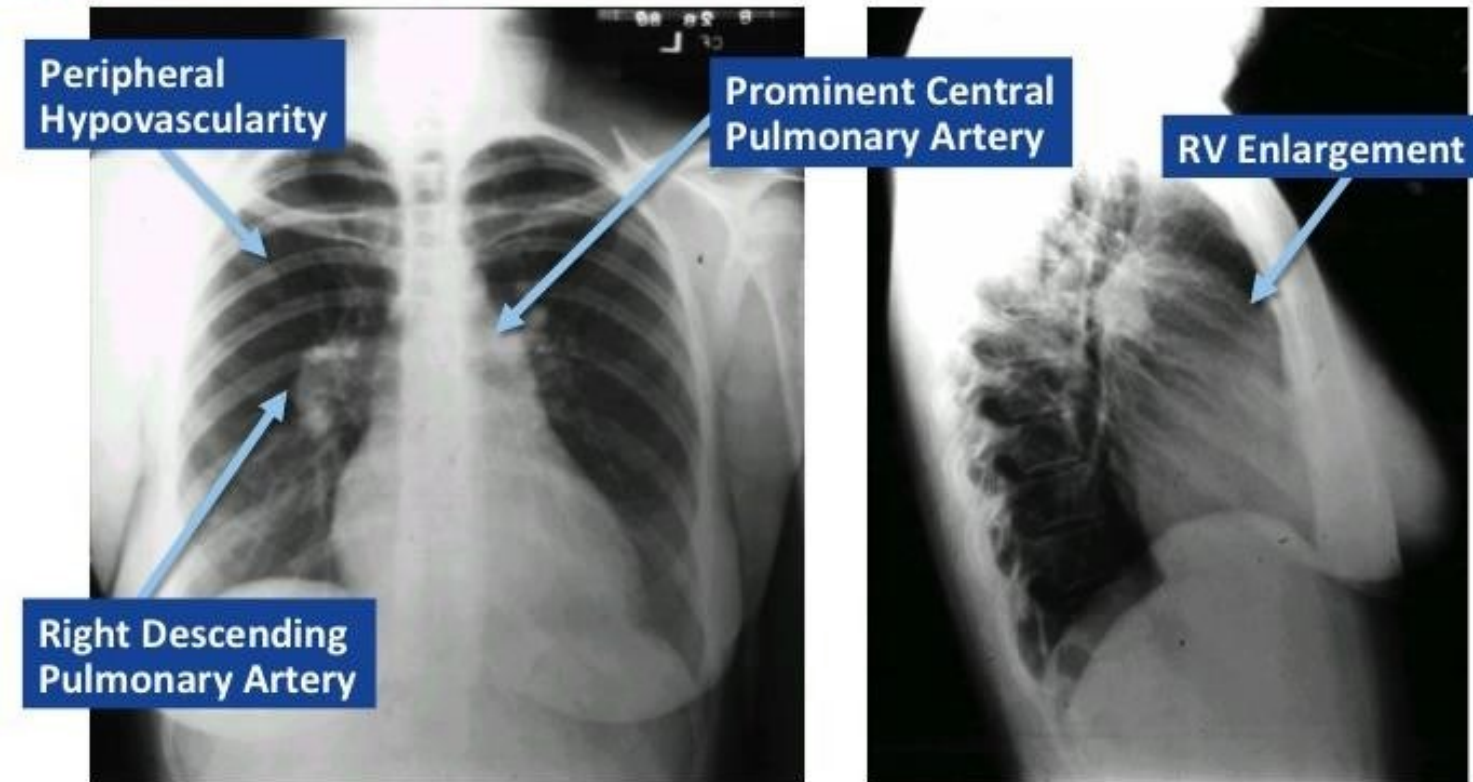


Pulmonary Hypertension

Diagnostics:

CXR shows enlarged pulmonary arteries with or without clear lung fields based on the cause of pulmonary hypertension

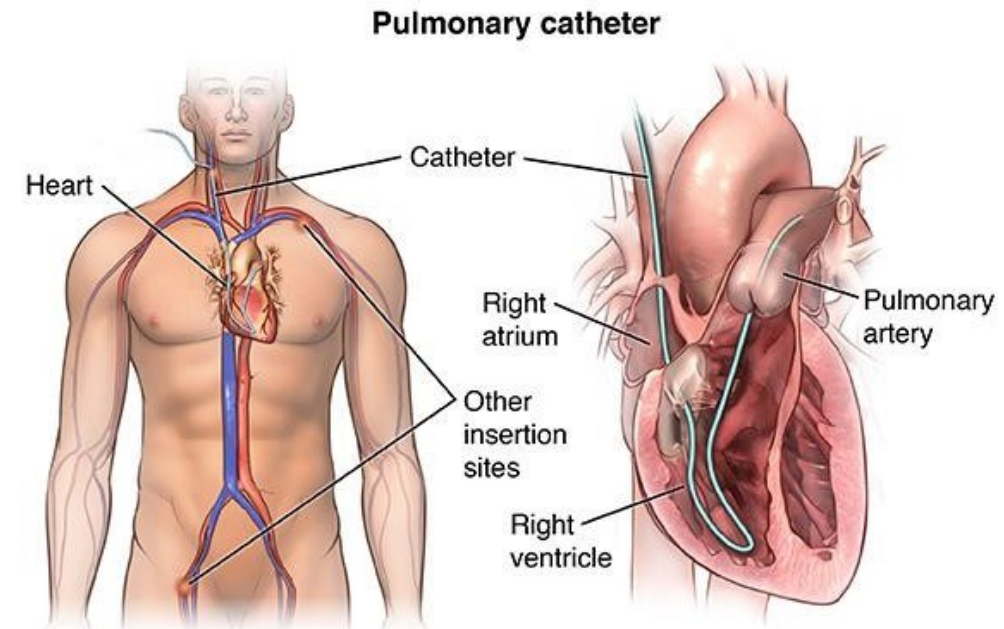
Figure 1. Characteristic X-Ray of a Patient with PAH



Pulmonary Hypertension

Diagnostics:

- Echocardiogram
 - Dilated pulmonary artery
 - Dilatation/hypertrophy of RA and RV
 - Abnormal movement of IV septum (due to increased right ventricular volume)
- Right heart catheterization: required for confirmatory diagnosis



Pulmonary Hypertension

Management:

Supportive:

- Oxygen therapy for patients with COPD and a $\text{PaO}_2 < 60$ mm Hg
- Diuretics for volume overload
- If the pulmonary HTN is secondary to another disease process (e.g., recurrent PE, HF), then the underlying disease should be treated and optimized.
- Vasoactive agents are typically used in PAH (Group 1)

Calcium channel blockers	e.g., Amlodipine
Endothelin receptor antagonists	Ambrisentan
	Bosentan
	Macitentan
Phosphodiesterase type 5 inhibitors (NO pathway)	Sildenafil Tadalafil
Guanylate cyclase stimulators (NO pathway)	Riociguat
Prostacyclin analogues	Epoprostenol (intravenous)
	Iloprost (inhaled)
	Treprostinil (subcutaneous)

Summary

Pulmonary Embolism (PE) features

mnemonic : DEAD SUDDENLY

- D** Dyspnea
- E** Elevated D-dimer
- A** Acute chest pain
- D** Dead if missed
- S** Spiral CT to diagnose
- U** Unilateral leg swelling
- D** DVT source
- D** Drops SpO₂
- E** ECG: S1Q3T3 (rare)
- N** Needs anticoagulation fast
- L** Lungs clear
- Y** You better not miss it



Case Scenario 1

Two days after undergoing a left total hip replacement, a 68-year-old man has sudden onset of shortness of breath and chest pain. He has type 2 diabetes mellitus and bilateral osteoarthritis of the hips. Prior to admission to the hospital, his medications were metformin and naproxen. His temperature is 37.8°C (100.0°F), pulse is 110/min, respirations are 30/min, and blood pressure is 106/70 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 89%. The lungs are clear to auscultation. Cardiac examination shows no abnormalities. The right lower extremity is swollen. Right foot flexion in an upward direction causes pain in the right calf. Pedal pulses are palpable. Supplemental oxygen therapy and intravenous fluid resuscitation are begun. His hematocrit is 30% and serum creatinine concentration is 1.2 mg/dL. Arterial blood gas analysis on room air is shown. An ECG shows sinus tachycardia and right axis deviation. In addition to intravenous anticoagulation, which of the following is the most appropriate next step in management?

- a. Perform compression ultrasonography
- b. Administer alteplase
- c. Obtain a spiral CT angiography
- d. Measure D-dimer levels
- e. Obtain an x-ray of the chest
- f. Perform an echocardiography
- g. Obtain a ventilation-perfusion scan

pH	7.48
pCO ₂	27 mm Hg
pO ₂	68 mm Hg
HCO ₃ ⁻	23 mEq/L

Case Scenario 2

A 67-year-old woman comes to the physician because of progressive swelling of her legs, shortness of breath on exertion, and abdominal discomfort for the past 3 months. She has also had nausea and decreased appetite, resulting in a 6-kg (13-lb) weight loss. She is being treated with paclitaxel for ovarian cancer and has had 2 episodes of pulmonary embolism during the past year. The patient has smoked one pack of cigarettes daily for 25 years and drinks two glasses of wine daily. She appears tired. The patient is 164 cm (5 ft 4 in) tall and weighs 54 kg (119 lb); BMI is 20 kg/m². Temperature is 37.0°C (98.6°F), pulse is 115/min, respirations are 18/min, and blood pressure is 135/85 mm Hg. Examination shows conjunctival icterus, protuberant abdomen with a fluid wave and shifting dullness, and 2+ pitting edema of the lower extremities. Lungs are clear to auscultation. There is visible distention of the jugular veins. Which of the following is the most likely explanation for this patient's findings?

- a. Liver metastases
- b. Drug toxicity
- c. Right ventricular failure
- d. Liver cirrhosis
- e. Obstruction of the hepatic veins

Questions?

ARE YOU A PULMONARY EMBOLISM ?
CAUSE YOU TAKE MY BREATH AWAY



References & further reading

- Step-up to medicine 5th edition
- Harrison's Principles of Internal Medicine
20th edition
- Amboss
- Cecil Essentials of Medicine 10th edition
- First Aid for the internal medicine board
4th edition
- Osmosis

Thank You